
EMCH 574 – Vibration and Waves

Homework 03

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Due date TBD

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Assignment

- G signifies students that take this class for Graduate Credit; they must do all the items.
- UG students are not required to do the G work, but they may do it for bonus points.
- Problems solved beyond the minimum requirements will be given bonus points.
- Challenge problems are optional; they will earn bonus points.

Notes

- Show your work to obtain partial credit if appropriate.
- Always display input data!
- Partial credit might be given based on work done and MATLAB codes if they are attached; without them, no partial credit can be given.
- Display numerical results to three significant digits (four if the first digit is 1).
- Use SI system of measurement [reference](#).
- Scale units using appropriate prefixes (e.g., nano, micro, milli, kilo, mega, giga, and so on) [reference](#).
- Problems solved beyond the minimum requirements will be given bonus points.
- Challenge problems are optional. They may be solved for extra credit.
- Normalized plots are plots with the *y*-axis scaled between -1 and $+1$. A plotted variable is normalized by dividing it by its maximum absolute value.

Notations and Abbreviations

- BVP = boundary value problem.
- EOM = equation of motion.
- FBD = free body diagram.
- FRF = frequency response function.
- IVP = initial value problem.
- LHS = left hand side.
- NLM = Newton's law of motion.
- ODE = ordinary differential equation.
- PDE = partial differential equation.
- RHS = right hand side.

1 String Vibration

A.1 General Analysis of String Vibration

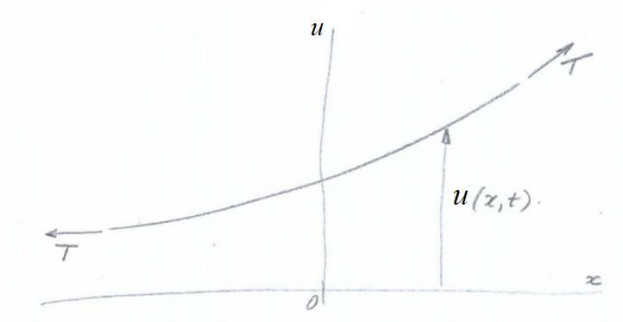


Figure 1: Taut string under tensile load T . (a) General view. (b) Infinitesimal element.

Consider a taut string of mass m per unit length stretched by a constant tension load T , as shown in Figure 1. The string undergoes small-amplitude transverse vibration with frequency ω . Its vibration motion is described by

$$u(x, t) = U(x)e^{-i\omega t}. \quad (1)$$

Do the following.

- (1) State the assumption made about the string tension T during the vibration motion u .
- (2) Draw free-body diagram (FBD) of an infinitesimal string element of length ds and write sum of y forces in terms of string tension T and slope α .
- (3) Apply small-displacement approximation and write the approximate expressions for ds , $\sin \alpha$, $\sin(\alpha + d\alpha)$.
- (4) Write the linearized expression for the sum of y forces.
- (5) Express the slope increment $d\alpha$ in terms of space increment dx and the relevant partial derivative of the displacement $u(x, t)$.
- (6) Apply Newton's law of motion (NLM) and write the NLM equation.
- (7) Write the governing equation for string vibration in terms of tension T , mass per unit length m , and the relevant partial derivatives of the displacement $u(x, t)$.
- (8) Define wavespeed c for wave traveling in a taut string in terms of tension T and mass per unit length m .
- (9) Write the wave equation.
- (10) Substitute Equation (1) into item (9) and eliminate the time dependency $e^{-i\omega t}$ to obtain an ODE defining $U(x)$.
- (11) Recall the definition of the wavenumber γ and substitute into (10) to obtain an ODE in terms of only γ and $U(x)$.
- (12) Assume exponential solution of the form $U(x) = e^{sx}$.

- (13) Develop the characteristic equation for s .
- (14) Solve the characteristic equation and write the roots s_1, s_2 .
- (15) Substitute (14) into (12) and write the general solution for $U(x)$ in terms of constants A_1, A_2 and complex exponentials depending on γ .
- (16) Use Euler's formulae to rewrite (15) in terms of new constants C_1, C_2 and trigonometric functions depending on γ .
- (17) Substitute (16) into Equation (1) and write the general solution for $u(x, t)$ in terms of constants C_1 and C_2 , trigonometric functions of γx , and a complex exponential function of ωt .
- (18) Graduate work. Write the general solution for $u(x, t)$ in terms of complex exponential functions of γ and ω .

A.2 Analyze Natural Frequencies and Modeshapes of a Vibrating String

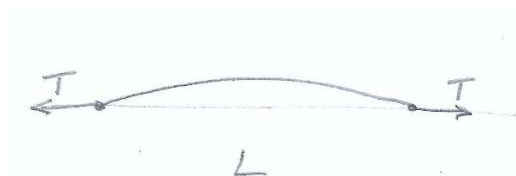


Figure 2: Taut string of length L under constant tension T .

Consider a taut string of length L and mass m per unit length. The string is pinned at both ends and stretched by a constant tension load T , as shown in Figure 2. The string undergoes transverse vibration with frequency ω .

Do the following.

- (1) Write the boundary conditions at the two ends of string.
- (2) Recall from Section A.1 the general solution for $u(x, t)$, expressed as the product of two terms.
 - (a) First term: a function $U(x)$ depending on constants C_1, C_2 and trigonometric functions of γx .
 - (b) Second term: a complex exponential function of ωt .
- (3) Substitute (2) into (1) and write boundary conditions in terms of constants C_1, C_2 .
- (4) Solve (3) for C_2 and obtain an equation for C_1 .
- (5) Discuss the trivial solution.
- (6) Choose to search for a nontrivial solution and define the equation that needs to be solved to find a nontrivial solution.
- (7) Introduce the notation $z = \gamma L$ and write the eigenvalue equation in terms of z .
- (8) Solve the eigenvalue equation and write the general expression for the eigenvalues $z_j, j = 1, 2, 3, \dots$.
- (9) Use (8) to write the general expression for the wavenumber γ_j in terms of j and length L .
- (10) Recall from HW01 the definition of wavenumber γ in terms of frequency ω and wavespeed c .
- (11) Solve (10) for frequency ω in terms of wavenumber γ and wavespeed c .
- (12) Use (9) and (11) to write the general expression of the natural frequency ω_j in terms of j , wavespeed c , and string length L .
- (13) Use (12) to write the general expression of the Hz frequency f_j in terms of j , wavespeed c , and string length L .
- (14) Explain the meaning of fundamental frequency and overtones.
- (15) Recall from HW01 the relationship between wavelength λ and wavenumber γ .
- (16) Substitute (9) into (15) to write the general expression for the wavelength λ_j in terms of j and string length L .

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- (17) Use (16) to explain which wavelength multiples could accommodate vibratory standing waves in a string of length L fixed at both ends.
- (18) Recall from (2)(a) the expression for $U(x)$.
- (19) Use (4) to eliminate C_2 .
- (20) Set $C_1 = 1$ in item (19) and write the general expression for the modeshape $U_j(x)$ in terms of the wavenumber γ_j .
- (21) Give the expression of modeshape orthonormality.

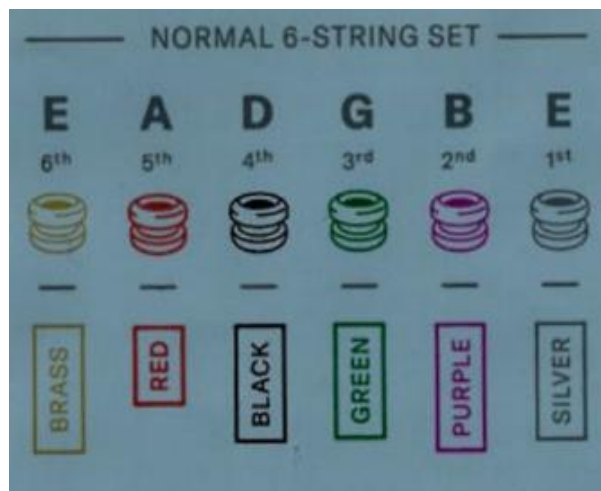
A.3 Calculate String Vibration

Consider a taut string of length $L = 22$ in tuned to the note G3. The purpose is to calculate the string vibration characteristics and modeshapes for $j = 1, \dots, N$ with $N = 5$. Verify mode orthonormality using numerical integration on `linspace(0, L, Nx)` with $N_x = 1000$. Recall the appropriate results from class instruction and the Blackboard notes.

Do the following.

- (1) Display input data.
- (2) State the note symbol (letter and octave) and the corresponding fundamental Hz frequency f_1 of the string.
- (3) Use frequency f_1 and length L to calculate the wavespeed c .
- (4) Calculate and display modal wavenumbers γ_j and frequencies ω_j .
- (5) Calculate the modeshapes, plot each mode separately, and then overlay all N modes in a single plot.
- (6) Verify modeshape orthonormality by numerical integration.
- (7) Discuss your results.

A.4 Calculate D'Addario Guitar Strings



String color	Brass	Red	Black	Green	Purple	Silver
Musical note	E	A	D	G	B	E
M (g)	6.42	4.18	2.45	1.332	0.655	0.367
T (lbf)	24.95	28.93	29.93	30.05	23.31	23.36

Table 1: *D'Addario light-gauge guitar string data.*

Consider a set of six light-gauge D'Addario guitar strings. All six strings have length $L = 25.0$ in. The mass M and tension T of each string are listed in Table 1.

(1) Display the input data.

For each of the six strings, complete items (2)–(11).

(2) Display the string length L in meters.

(3) Calculate and display the distributed mass m in kg/m.

(4) Display the tension T in newtons.

(5) Calculate the wavespeed c in m/s.

(6) Calculate the fundamental frequency f_1 in hertz.

(7) Find the musical note that most closely approximates the frequency and give its letter name and octave number (e.g., G3 signifies note G in the third octave).

(8) Display the frequency of the musical note found in item (7).

(9) Calculate the percentage difference between your result and the nominal frequency of that musical note.

(10) Comment on your findings and identify possible sources of error.

(11) Graduate work. Calculate the necessary adjustments in mass, tension, or both to achieve a frequency match within 0.00%.

A.5 Calculate Vibration of a Plucked String

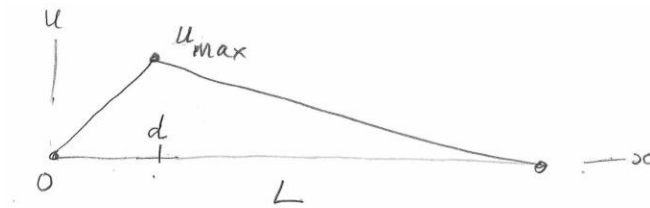


Figure 3: Taut string plucked at an arbitrary location d .

Consider the taut string shown in Figure 3. It has length $L = 25.5$ in and is tuned to the note G3. The string is plucked by applying an initial displacement $u_{\max} = 1.25$ mm at a location d . Consider three plucking locations. They are $d = L/2$, $d = L/4$, and $d = 7L/10$. Calculate the response $u(x, t)$ using the modal expansion method with $N = 10$. Recall the appropriate results from class instruction and the Blackboard notes.

Do the following.

- (1) Display input data.
- (2) State the note symbol (letter and octave) and the corresponding fundamental Hz frequency f_1 of the string. Calculate the corresponding fundamental mode period τ_1 .
- (3) Use frequency f_1 and length L to calculate the wavespeed c .
- (4) Calculate modal wavenumbers γ_j and frequencies ω_j .
- (5) Calculate the N modes and plot them together.
- (6) For each plucking location, complete items (a)–(d).
 - (a) Display the value d and plot the initial shape of the string $u_0(x)$.
 - (b) Calculate by numerical integration the modal participation factors A_j , display A_j in mm, and do a bar plot.
 - (c) Plot the time response of the string at the plucking location $u(d, t)$ for $t = 0, \dots, t_{\max}$ with $t_{\max} = 3\tau_1$.
 - (d) Graduate work. Reconstruct the initial string shape and plot it overlaid on the actual shape.
- (7) Compare and discuss the results for the three plucking locations.

A.6 Calculate Frequency Response of a Struck String

This problem addresses the frequency response of a taut string when struck by a point load like in the case of the hammered dulcimer (Figure 4).

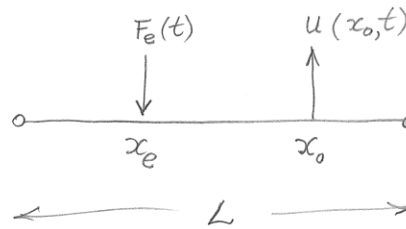


Figure 4: Struck string. (a) The hammered dulcimer. (b) Schematic.

Consider a taut string of length $L = 25.5$ in and distributed mass $m = 0.0015$ kg/m tuned to the note G3. Assume the same damping ratio, $\zeta_0 = 2.1\%$, in every mode. The string is struck at location x_e , and its response is measured with a piezoelectric acoustic pickup at location x_0 . Consider the following three cases.

Case	x_e	x_0
(a)	$L/3$	$L/3$
(b)	$L/3$	$L/2$
(c)	$L/2$	$L/2$

The striking force is harmonic with amplitude $F_e = 0.5$ N over the frequency range $f_{\text{start}} = 20$ Hz through $f_{\text{end}} = 2000$ Hz, using $N_f = 5000$ steps. Calculate and plot the frequency response function (FRF) over this range using modal expansion with $N = 10$ modes.

Do the following.

- (1) Display input data.
- (2) State the note name and octave and the corresponding fundamental Hz frequency f_1 of the string. Calculate the corresponding period τ_1 and the wavespeed c .
- (3) For $j = 1, \dots, N$, calculate and display the modal wavenumber γ_j , frequency ω_j , mass m_j , and damping ratio ζ_j . Plot the modeshapes together.
- (4) Define the frequency range f using `linspace( $f_{\text{start}}$ ,  $f_{\text{end}}$ ,  $N_f$ )` and then define $\omega = 2\pi f$.

-
- (5) For each (x_e, x_0) case, plot $|\text{FRF}(\omega; x_e, x_0)|$ in decibels and find the frequencies, nearest musical notes, and corresponding string-mode indices.
- (6) Graduate work. Plot the time response of the normalized sound picked up at $x_0 = L/2$ when the string is struck at $x_e = L/3$. Use the dominant note frequency from the FRF plot as the excitation frequency, and cover three periods of that frequency.

2 Flexural Vibration

B.1 General Analysis of Flexural Vibration

Consider a uniform beam of mass m per unit length and bending stiffness EI undergoing flexural motion with vertical displacement $w(x, t)$ as shown in Figure 5. Centroidal axes are assumed.

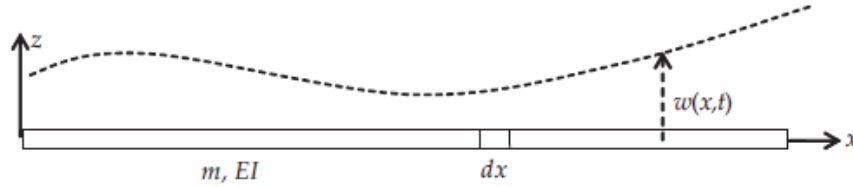


Figure 5: Beam undergoing flexural vibration.

Do the following.

- (1) State the assumptions needed for performing the flexural vibration analysis.
- (2) Draw free-body diagram (FBD) of an infinitesimal beam element dx and show the stress resultants N , M , V .
- (3) Write the definitions of the stress resultants N and M .
- (4) Use stress-strain relation to write the stress resultants in terms of strains.
- (5) Perform kinematic analysis to express the stress resultant M in terms of partial derivatives of the displacement w .
- (6) Perform free-body analysis of the infinitesimal element dx and find the sum of forces ΣF_z and sum of moments ΣM .
- (7) Apply Newton's law of motion (NLM) in the z direction and obtain an equation for the shear force V .
- (8) Apply NLM in the rotational direction and obtain an equation for the moment M .
- (9) Substitute (6) into (8) and get an equation that relates M and V .
- (10) Substitute (5) into (9) and obtain the governing equation.
- (11) Define the flexural constant a in terms of E , I , m .
- (12) Substitute (11) into (10) to obtain the governing equation as a PDE in x , t , and flexural constant a .
- (13) Assume harmonic vibration of the form $w(x, t) = W(x)e^{-i\omega t}$.
- (14) Substitute (13) into (12) and obtain an ODE in x with constants a , ω .
- (15) Define the wavenumber γ in terms of ω and a and substitute in (14) to write the ODE with only one constant, γ .
- (16) Assume $W(x)$ is an exponential function of x of the form $W(x) = e^{sx}$.
- (17) Substitute (16) into (15) and derive the characteristic equation.

- (18) Find the four roots of the characteristic equation.
- (19) Write the general solution $W(x)$ in terms of constants A_1, A_2, A_3, A_4 and complex exponential functions of γx .
- (20) Rewrite the general solution (19) in terms of constants C_1, C_2, C_3 , and C_4 multiplying trigonometric and hyperbolic functions of γx .
- (21) Graduate work. Explain how one can find the constants C_1, C_2, C_3 , and C_4 .

B.2 Analyze Cantilever Beam Vibration

A cantilever beam is fixed at one end and free at the other end. Consider a cantilever beam of length L as shown in Figure 6. The beam has mass density per unit length m , elastic modulus E , and cross-section second moment of area I .

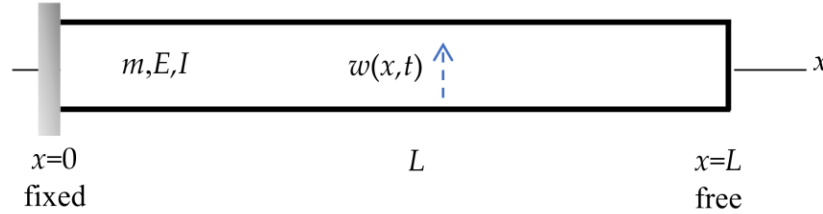


Figure 6: Vibration of a cantilever beam of length L . The cantilever beam is fixed at one end and free at the other end.

Do the following.

- (1) Write the boundary conditions at the two ends of the beam.
- (2) Recall from the notes the general solution $w(x, t)$, expressed as the product of two terms.
 - (a) First term: a function $W(x)$ depending on constants C_1, C_2, C_3, C_4 , and trigonometric and hyperbolic functions of γx .
 - (b) Second term: a complex exponential function of ωt .
- (3) Substitute (2) into (1) and write boundary conditions in terms of constants C_1, C_2, C_3, C_4 .
- (4) Solve (3) for C_3, C_4 in terms of C_1, C_2 and obtain a set of equations for C_1, C_2 .
- (5) Discuss the trivial solution.
- (6) Choose to search for a nontrivial solution and define the equation that needs to be solved to find a nontrivial solution.
- (7) Introduce the notation $z = \gamma L$ and write the transcendental eigenvalue equation in terms of z .
- (8) Use the MATLAB function `fzero` to solve the transcendental eigenvalue equation and find z_j , $j = 1, \dots, N$, with $N = 5$.
- (9) Use item (7) to write the general expression for the wavenumber γ_j in terms of z_j and length L .
- (10) Recall from Problem B.1 the definition of wavenumber γ in terms of frequency ω and flexural constant a .
- (11) Solve (10) for frequency ω in terms of wavenumber γ and flexural constant a .
- (12) Use (9) and (11) to write the general expression of the natural frequency ω_j in terms of eigenvalue z_j , beam constant a , and beam length L .
- (13) Recall the definition of a in terms of bending stiffness EI and mass per unit length m , and obtain expressions for the natural frequencies ω_j and f_j in terms of z_j, EI, m , and L .
- (14) Use (4) to obtain an expression for the ratio C_1/C_2 in terms of wavenumber γ and beam

length L . Denote this expression to be $-\beta$.

- (15) Recall from (2)(a) the expression for $W(x)$.
- (16) Use (4) and (14) to write the constants C_1 , C_3 , C_4 in terms of C_2 .
- (17) Substitute (16) into (15) and write $W(x)$ in terms of C_2 , γx , β .
- (18) Set $C_2 = 1$ and write the modeshapes $W_j(x)$ in terms of γ_j and β_j .
- (19) Give the expression of modeshape orthonormality.
- (20) Graduate work. Complete items (a) and (b).
 - (a) Write the approximate expression for the eigenvalues z_j with $j \geq 6$.
 - (b) Solve the transcendental equation at item (7) for $N = 10$ and compare with the prediction using the approximate expression of item (a).

B.3 Calculate Cantilever Beam Vibration

Assume a cantilever beam of length $L = 445 \text{ mm}$ and cross-section dimensions $h = 3 \text{ mm}$, $b = 37 \text{ mm}$. The material is steel with $E = 200 \text{ GPa}$ and $\rho = 7850 \text{ kg/m}^3$.

Do the following.

- (1) Display input data.
- (2) Calculate the cross-section area A .
- (3) Calculate mass per unit length m .
- (4) Calculate second moment of area I .
- (5) Calculate flexural stiffness EI .
- (6) Calculate flexural constant a .
- (7) Solve the transcendental equation defined in Problem B.2 item (7) to find z_j , $j = 1, \dots, N$ with $N = 5$.
- (8) Calculate the wavenumbers γ_j .
- (9) Calculate angular frequencies ω_j .
- (10) Calculate Hz frequencies f_j .
- (11) Calculate the modeshape coefficients β_j .
- (12) Calculate the modeshapes $W_j(x)$, plot each mode separately, and then plot all N modes together.
- (13) Verify modeshape orthonormality by numerical integration.
- (14) Comment on your results.

B.4 Calculate the 512 C Tuning Fork

Consider the 512 C aluminum-alloy tuning fork shown in Figure 7. Common values for aluminum elastic modulus and density are $E = 70 \text{ GPa}$ and $\rho = 2700 \text{ kg/m}^3$. The tuning fork is intended to produce the middle-C note C_5 on the scientific pitch scale, with fundamental frequency $f_1 = 512 \text{ Hz}$.

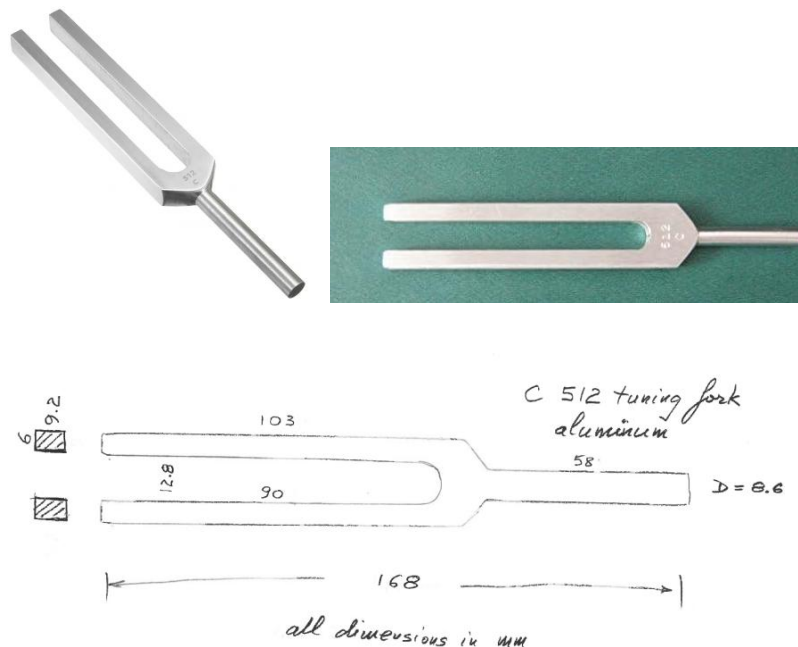


Figure 7: 512 C tuning fork (product reference). The measured sketch was prepared from the physical fork.

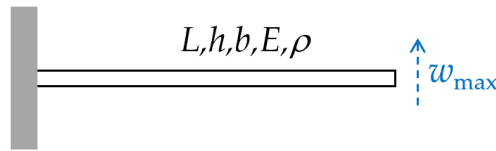
Use the cantilever-beam analysis developed in Problem B.2 to perform the engineering evaluation of this tuning fork.

Do the following.

- (1) Display input data.
- (2) Use the sketch in Figure 7 to estimate the cantilever-beam length L . Hint: use engineering intuition to estimate where the root of the cantilever beam could be assumed.
- (3) Use the sketch in Figure 7 to obtain the cantilever-beam cross-section dimensions h and b .
- (4) Calculate cross-section area A .
- (5) Calculate mass per unit length m .
- (6) Calculate second moment of area I .
- (7) Calculate flexural stiffness EI .
- (8) Calculate flexural constant a .
- (9) Use the results of Problem B.3 to calculate the fundamental frequency f_1 in Hz.
- (10) Compare the calculated f_1 with the nominal tuning fork frequency of 512 Hz, calculate the percentage difference, and comment on your results.

- (11) Manually adjust the estimated beam length to a new value L_{new} until the new frequency $f_{1,\text{new}}$ equals 512 Hz.
- (12) Comment on the engineering estimation and adjustment process developed in the solution of this problem.
- (13) Graduate work. Find the frequencies of the next three higher harmonics of the tuning fork after length adjustment.

B.5 Calculate Vibration of a Plucked Reed



Cantilever-reed model and initial tip displacement.

Consider a cantilever beam of length $L = 14 \text{ mm}$ and cross-section dimensions $h = 0.3 \text{ mm}$ and $b = 0.75 \text{ mm}$. The material is steel with $E = 200 \text{ GPa}$ and $\rho = 7850 \text{ kg/m}^3$. The beam is plucked with an initial tip displacement $w_{\max} = 0.15 \text{ mm}$. Calculate the response $w(x, t)$ using the modal expansion method. Recall the appropriate results from class and the Blackboard notes.

Do the following.

- (1) Display input data.
- (2) Calculate the cross-section area A , mass per unit length m , cross-section second moment of area I , flexural stiffness EI , and flexural constant a .
- (3) Calculate the roots of the cantilever eigenvalue equation for $j = 1, \dots, N$ with $N = 5$.
- (4) Calculate the fundamental Hz frequency f_1 and state the symbol (letter and octave) of the nearest musical note. Calculate the fundamental mode period τ_1 .
- (5) Recall the formulas for cantilever beam modeshapes and plot the modes $j = 1, \dots, N$.
- (6) Calculate and plot the initial deformed shape of the beam $w_0(x)$.
- (7) Assume the modal expansion method, calculate the modal coefficients A_j , $j = 1, \dots, N$, and do a bar plot.
- (8) Plot the time response of the beam tip $w(L, t)$ for three fundamental-mode periods.
- (9) Comment on your results.

3 Extra Credit Challenge Problems

Challenge problems are optional.

C.1 Analyze Flexural Vibration in a Free-Free Beam

- (1) Sketch a free-free beam indicating all the information needed for the analysis.
- (2) Write the boundary conditions at the two ends of the beam in terms of displacement $w(x, t)$ and stress resultants $M(x, t)$, $V(x, t)$.
- (3) Repeat all the steps of Section B.2.

C.2 Calculate Wind-Chime Tones

Consider the wind chime shown in Figure 8 and Figure 9.



Figure 8: *Wind chime.*

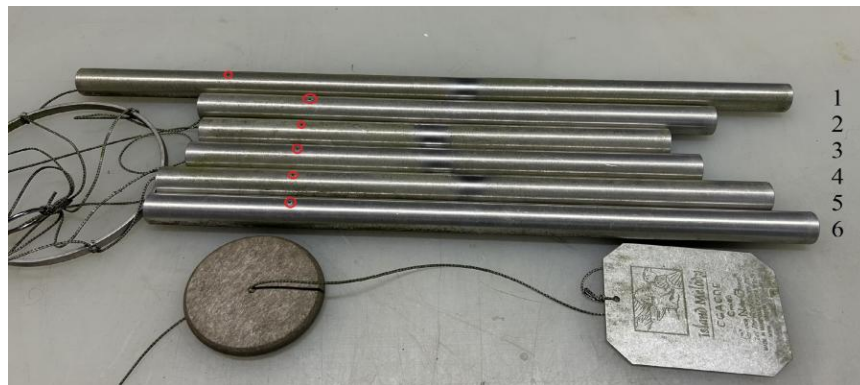


Figure 9: *Tubes of the wind chime with the suspension points circled in red.*

The wind chime tubes are aluminum with $E = 70 \text{ GPa}$ and $\rho = 2700 \text{ kg/m}^3$. Their outer and inner diameters are $D = 22.3 \text{ mm}$ and $d = 19.4 \text{ mm}$, respectively. Each tube has length L and is suspended a distance e from one end. The measured values are listed in Table 2.

Tube	L (mm)	e (mm)
1	500	110
2	352	78
3	312	70
4	332	74
5	384	87
6	406	90

Table 2: *Measured lengths and suspension distances for the six wind chime tubes.*

Use the free-free beam analysis developed in Problem C.1 to perform the engineering evaluation of the wind chime tubes.

Do the following.

- (1) Calculate the cross-section area A .
- (2) Calculate mass per unit length m .
- (3) Calculate second moment of area I .
- (4) Calculate flexural stiffness EI .
- (5) Calculate flexural constant a .
- (6) Use the MATLAB function `fzero` to find the first root of the transcendental eigenvalue equation defined in Problem C.1.
- (7) Use the length values in Table 2 to calculate the wavenumber γ and corresponding frequencies ω_1 and f_1 for each tube.
- (8) Use [reference](#) to estimate the musical notes that might be associated with the frequencies of the six tubes.
- (9) Graduate work. Complete items (a) and (b).
 - (a) Calculate and plot the modeshape $W_1(x)$ for each of the six tubes.
 - (b) Estimate the location x_0 where the modeshape $W_1(x)$ crosses zero. Compare x_0 with the values of e in Table 2, calculate the percentage difference, and comment on the results.
- (10) Graduate work. Use manual iteration to find adjusted tube lengths that produce exact natural notes, without sharps, for item (8).

C.3 Calculate Flexural Vibration in a Free-Free Beam

Assume a free-free beam of length $L = 445 \text{ mm}$ and cross-section dimensions $h = 3 \text{ mm}$ and $b = 37 \text{ mm}$. The material is steel with $E = 200 \text{ GPa}$ and $\rho = 7850 \text{ kg/m}^3$.

Do the following.

- (1) Repeat the steps of Section B.3.

C.4 Analyze Flexural Vibration of a Simply Supported Beam

- (1) Sketch a simply supported beam indicating all the information needed for the analysis.
- (2) Write the boundary conditions at the two ends of the beam in terms of displacement $w(x, t)$ and stress resultants $M(x, t)$, $V(x, t)$.
- (3) Recall from the class notes the general solution $w(x, t)$, expressed as the product of two terms.
 - (a) First term: a function $W(x)$ depending on constants C_1 , C_2 , C_3 , C_4 , and trigonometric and hyperbolic functions of γx .
 - (b) Second term: a complex exponential function of ωt .
- (4) Substitute (3) into (2) and write boundary conditions in terms of constants C_1 , C_2 , C_3 , C_4 .
- (5) Solve (4) for C_2 , C_4 and obtain a set of equations for C_1 , C_3 .
- (6) Discuss the trivial solution.
- (7) Choose to search for the nontrivial solution and define the equation that needs to be solved to find a nontrivial solution.
- (8) Introduce the notation $z = \gamma L$ and write the eigenvalue equation in terms of z .
- (9) Write the general expression of the roots z_j of the eigenvalue equation.
- (10) Use item (8) to write the general expression for the wavenumber γ_j in terms of j and length L .
- (11) Recall from class notes the definition of wavenumber γ in terms of frequency ω and flexural constant a .
- (12) Solve (11) for frequency ω in terms of wavenumber γ and flexural constant a .
- (13) Use (10) and (12) to write the general expression of the natural frequency ω_j in terms of j , beam constant a , and beam length L .
- (14) Recall the definition of a in terms of bending stiffness EI and mass per unit length m , and obtain expressions for the natural frequencies ω_j and f_j in terms of j , EI , m , and L .
- (15) Use item (5) to solve for C_3 .
- (16) Substitute the values found so far for C_2 , C_3 , C_4 and write $W(x)$ in terms of C_1 and γx .
- (17) Set $C_1 = 1$ and write the modeshapes $W_j(x)$ in terms of γ_j .
- (18) Give the expression of modeshape orthonormality.

C.5 Calculate Simply Supported Flexural Vibration

Assume a simply supported beam of length $L = 445 \text{ mm}$ and cross-section dimensions $h = 3 \text{ mm}$ and $b = 37 \text{ mm}$. The material is steel with $E = 200 \text{ GPa}$ and $\rho = 7850 \text{ kg/m}^3$.

Do the following.

- (1) Repeat the steps of Section B.3.

C.6 Analysis of the Wave Equation in a Slender Bar

Assume a uniform slender bar of cross section A and density ρ (Figure 10). The bar undergoes time-varying axial displacement $u(x, t)$. The displacement is assumed uniform across the bar cross section.

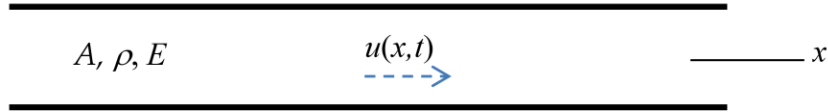


Figure 10: Wave motion in a uniform slender bar.

Do the following.

- (1) State the assumption made about motion and deformation in the slender bar.
- (2) Draw free-body diagram (FBD) of an infinitesimal bar element of length dx and write Newton's law of motion (NLM) equation.
- (3) Write elasticity equation that relates stress and displacement.
- (4) Substitute (3) into (2) and derive the partial differential equation (PDE) in terms of displacement.
- (5) Define the wavespeed c in terms of elasticity modulus E and density ρ .
- (6) Write the wave equation.
- (7) Assume separation of x and t variables and write the solution as a product between an x function $U(x)$ and an exponential function in t of the form $e^{-i\omega t}$.
- (8) Substitute (7) into (6) and derive an ordinary differential equation (ODE) in terms of c , ω , $U(x)$.
- (9) Recall from HW01 the definition of wavenumber γ in terms of frequency ω and wavespeed c and use it to simplify (8).
- (10) Assume $U(x)$ as an exponential function of x of the form e^{sx} .
- (11) Substitute (10) into (9) and derive the characteristic equation.
- (12) Find the roots s_1 , s_2 of the characteristic equation in terms of the wavenumber γ and write the general solution $U(x)$ in terms of complex exponentials of γx .
- (13) Add the time dependency defined at (7) and write the general solution for $u(x, t)$ in terms of complex exponentials of γx , ωt , and constants A_1 , A_2 .
- (14) Rewrite the general solution (13) in terms of constants C_1 , C_2 , trigonometric functions of γx and a complex exponential function of ωt .
- (15) Graduate work. Identify the forward and backward waves in the general solution derived in item (13).

C.7 Calculate Axial Vibration with Various Boundary Conditions

Consider a slender aluminum bar of diameter $D = 20 \text{ mm}$ and length $L = 3.0 \text{ m}$. Take the aluminum elastic modulus and density as $E = 70 \text{ GPa}$ and $\rho = 2700 \text{ kg/m}^3$. The bar undergoes axial vibration with frequency ω . Calculate the bar vibration characteristics and modeshapes for $j = 1, \dots, N$ with $N = 5$, and verify mode orthonormality using numerical integration on `linspace(0, L, N_x)` with $N_x = 1000$. Consider the following boundary conditions.

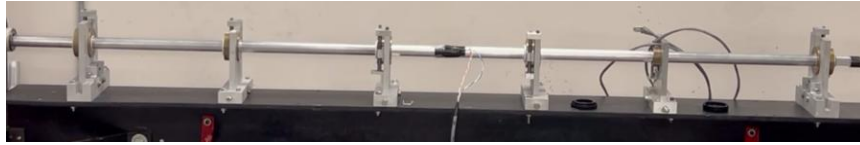
Case	$x = 0$	$x = L$
(a)	Fixed	Fixed
(b)	Fixed	Free
(c)	Free	Free

- (1) Display the input data.
- (2) Calculate the wavespeed c .

For each of the three boundary-condition cases, complete items (3)–(8).

- (3) Display boundary conditions.
- (4) Calculate and display modal vibration characteristics.
- (5) Calculate the displacement modeshapes and plot them together.
- (6) Verify modeshape orthonormality by numerical integration.
- (7) Calculate the normalized stress modeshapes and plot them together.
- (8) Discuss your results.

C.8 Calculate the Hopkinson Bar



Split-Hopkinson pressure bar apparatus (reference).

The Hopkinson pressure bar is used to create and measure axial stress pulses. It consists of a long and slender metallic bar which is hit at one end. The hit creates a stress pulse that propagates in the bar and is measured by strain gauges. A common application is to measure the compression behavior of a small specimen placed in what is called the Split-Hopkinson bar. Consider an aluminum Hopkinson pressure bar of diameter $D = 20 \text{ mm}$. Common values for aluminum elastic modulus and density are $E = 70 \text{ GPa}$ and $\rho = 2700 \text{ kg/m}^3$.

Do the following.

- (1) Calculate the axial wavespeed c in aluminum.
- (2) Assume the bar length is $L = 3.2 \text{ m}$ and calculate the time t required to travel the length of the bar.
- (3) Calculate the natural frequencies f_1, f_2, f_3 of the first three modes of bar axial vibration.
- (4) Calculate and plot the corresponding axial vibration modeshapes.
- (5) Graduate work. Research the split Hopkinson pressure bar and write a short description of the apparatus and its uses.

C.9 Analyze Axial Vibration in a Free-Free Bar

Consider a slender bar of finite length L as shown in Figure 11. The bar has cross-section area A , density ρ , elastic modulus E . The bar is free at both ends.

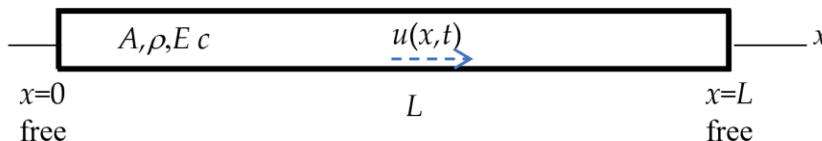


Figure 11: *Vibration of a slender bar of finite length L free at both ends.*

Do the following.

- (1) Write the boundary conditions at the two ends of the bar.
- (2) Recall from Problem C.6 the general solution $u(x, t)$ in terms of:
 - (a) First term: a function $U(x)$ depending on constants C_1, C_2 and trigonometric functions of γx .
 - (b) Second term: a complex exponential function of ωt .
- (3) Substitute (2) into (1) and write boundary conditions in terms of constants C_1, C_2 .
- (4) Solve (3) for C_1 and obtain an equation for C_2 .
- (5) Discuss the trivial solution.
- (6) Choose to search for a nontrivial solution and define the equation that needs to be solved to find a nontrivial solution.
- (7) Introduce the notation $z = \gamma L$ and write the eigenvalue equation in terms of z .
- (8) Solve the eigenvalue equation and write the general expression for the eigenvalues $z_j, j = 1, 2, 3, \dots$
- (9) Use (8) to write the general expression for the wavenumber γ_j in terms of j and length L .
- (10) Recall from HW01 the definition of wavenumber γ in terms of frequency ω and wavespeed c .
- (11) Use (10) to solve for frequency ω in terms of wavenumber γ and wavespeed c .
- (12) Use (9) and (11) to write the general expression of the natural frequency ω_j in terms of j , wavespeed c , and bar length L .
- (13) Use (12) to write the general expression of the Hz frequency f_j in terms of j , wavespeed c , and length L .
- (14) Explain the meaning of fundamental frequency and overtones.
- (15) Recall from HW01 the relationship between wavelength λ and wavenumber γ .
- (16) Substitute (9) into (15) to write the general expression for the wavelength λ_j in terms of j and length L .
- (17) Use (16) to explain which wavelength multiples can accommodate vibratory standing waves in

a bar of length L free at both ends.

- (18) Recall $U(x)$ from (2)(a) and use (4) to eliminate C_1 .
- (19) Set $C_2 = 1$ and write the general expression of the modeshape $U_j(x)$ in terms of the wavenumber γ_j .
- (20) Give the expression of modeshape orthonormality.
- (21) Write the general expression of the stress modeshapes.

C.10 Analyze Axial Vibration in a Fixed-Fixed Bar

Do the following.

- (1) Sketch a bar fixed at both ends indicating all the information needed for the analysis.
- (2) Write the boundary conditions at the two ends of the bar.
- (3) Repeat the steps of Problem C.9.

C.11 Analyze Axial Vibration in a Fixed-Free Bar

Do the following.

- (1)** Sketch a bar fixed at one end and free at the other end indicating all the information needed for the analysis.
- (2)** Write the boundary conditions at the two ends of the bar.
- (3)** Repeat the steps of Problem C.9.